# Mustafa Yanmaz

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## ABOUT ME

Junior Software Engineer focused on backend development, with hands-on experience in Spring Boot, ASP.NET Core, RESTful APIs, PostgreSQL, and microservice-based applications. Developed secure and scalable software solutions using Keycloak, JWT, RabbitMQ, Docker, and modern authentication and authorization practices. Completed two software development internships and built projects including a stock management microservice, a full-stack university information system, and a real-time IPv4/IPv6 intrusion detection and prevention system. Familiar with automated testing, CI/CD workflows, Linux-based development environments, and frontend technologies such as React and Next.js.

## EDUCATION AND TRAINING

04/09/2021 – 31/07/2026 Rize, Türkiye

**BACHELOR'S DEGREE IN COMPUTER ENGINEERING** Recep Tayyip Erdoğan University

## WORK EXPERIENCE

**SOFTWARE DEVELOPMENT INTERN – RECEP TAYYIP ERDOĞAN UNIVERSITY – DEPARTMENT OF INFORMATION TECHNOLOGIES** – 23/06/2025 – 02/09/2025 – RIZE, TÜRKIYE

- Developed and deployed a full-stack tea seedling monitoring platform using ASP.NET Core, PostgreSQL, React, and Tailwind CSS.
- Designed the relational data model and implemented JWT authentication, role-based authorization, advanced filtering, detailed seedling profiles, gallery management, and audit logging.
- Processed seedling data extracted from PDF catalogues using Python and collaborated with the university's IT Department on testing, error handling, and deployment.

**BACKEND DEVELOPMENT INTERN – AGITO SOFTWARE & CONSULTING** – 09/02/2026 – 22/05/2026 – İSTANBUL, TÜRKIYE

- Developed a stock management microservice using Spring Boot, layered architecture, RESTful APIs, DTOs, validation, object mapping, and global exception handling.
- Implemented product, stock, and sales operations, including FIFO-based stock deduction, stock-batch traceability, and average cost calculation.
- Integrated Keycloak and JWT for authentication and role-based authorization, RabbitMQ for low-stock email notifications, and Swagger/OpenAPI for API documentation.
- Contributed to application testing, debugging, code reviews, and team-based development workflows using Git and Docker.

## SKILLS

**Programming Languages**

Java | C | C++ | C# | Python | JavaScript

**Backend, Web Technologies & Databases**

Spring Boot | ASP.NET Core | Flask | Entity Framework Core | RESTful APIs | Keycloak | JWT | RabbitMQ | Swagger/OpenAPI | React | Next.js | Tailwind CSS | SQL | PostgreSQL | MySQL

**Development Tools, Testing & DevOps**

Git | GitHub | GitHub Actions | Docker | Linux | CMake | GoogleTest | OpenMP | MPI | PX4 | Gazebo

## PROJECTS

### Stock Management Microservice

- Developed a Spring Boot-based stock management microservice using layered architecture, RESTful APIs, DTOs, validation, mapping, and global exception handling.
- Implemented product, stock, and sales operations with FIFO-based stock deduction, stock-batch traceability, and average cost calculation.
- Integrated Keycloak/JWT authentication, RabbitMQ-based low-stock email notifications, and Swagger API documentation.

### ÇAYGEN – Tea Seedling Monitoring System

- Designed an SQL-based data model and processed seedling data extracted from PDF catalogues using Python.
- Implemented JWT authentication, role-based authorization, advanced filtering, detailed seedling profiles, gallery management, and audit logging.
- Built RESTful backend services with ASP.NET Core and PostgreSQL and developed the user interface using React and Tailwind CSS.

### Undergraduate Thesis — AI-Powered Real-Time IPv4/IPv6 IDS/IPS System

- Built an Orange Pi 5-based physical testbed to collect and label a custom IPv4/IPv6 network traffic dataset using normal traffic and 28 controlled attack scenarios.
- Evaluated five machine learning algorithms on six public IDS datasets, then tested the protocol-specific models on the custom IPv4/IPv6 dataset collected through the physical testbed using a two-stage binary and multi-class classification architecture.
- Developed an end-to-end real-time IDS/IPS prototype integrating one-second traffic analysis, threat scoring, automatic IPv4/IPv6 address blocking, and web-based visualization of detected security events, with processing times remaining below 50 ms during flood traffic tests.

### Book Exchange Platform (Parallel Computing)

- Developed a C++17 book exchange platform and parallelized search, genre analysis, and similarity matrix operations using OpenMP and MPI.
- Compared sequential and parallel implementations across multi-core and distributed environments to evaluate performance improvements.

### Distributed Swarm UAV System

- Developed and tested a multi-UAV simulation environment using PX4 and Gazebo for a distributed swarm UAV system.
- Configured the PX4- and Gazebo-based simulation environment and autonomous mission scenarios to support multi-UAV system testing and integration.
- Contributed to the testing of swarm coordination, autonomous missions, and communication between system components.

### Pet Care Reminder System

- Developed a Java-based pet care management and reminder application.
- Implemented core modules using linked lists, sparse matrices, stacks, queues, Heap Sort, Huffman coding, BFS/DFS, Hash Tables, B+ trees, KMP, and hashing techniques.
- Applied these data structures and algorithms to scheduling, data organization, compression, graph traversal, and efficient search operations.

### Cipher Master — Interactive Cryptography Learning Tool

- Developed an interactive application that visualizes the encryption and decryption processes of eight classical cryptographic algorithms in real time.

## LANGUAGE SKILLS

Other language(s): **ENGLISH — B2 (UPPER-INTERMEDIATE)**